Farshore: Arcade Trials

Phase 2 design packet / Bibek Pathak / October 8, 2026

A five-stop arcade voyage where each mythic rival uses a different game mechanic. Optional Llama planning chooses a legal rival strategy before a match.

Phase 2 designs document the current arcade prototype and explicitly labeled proposed controls. Maya, Daniel and Rosa are provisional Phase 1 persona sketches. Interview evidence and class peer critique have not been supplied.

Rubric	Included evidence
Architecture 20%	15-screen hierarchy, content strategy, A1 integration, storage and data flow
Flows 30%	Five persona-labeled tasks with control/recovery branches and eight edge cases
Wireframes 30%	28 annotated grayscale screens and major states, all five A1 model states
Spike 10%	Six real current-use-case calls, exact outputs and three failure records
Presentation 10%	12 timed slides with a full review appendix, 600-second notes and a separate contribution file

Phase 1 persona references

Provisional sketches guide tasks; arcade needs remain hypotheses

P1 Maya, 21

Goal: Win through skill and understand why the rival behaves differently.
Design need: Visible attack warnings, fixed scoring rules and a clear distinction between difficulty and rival strategy.
Flows: F2, F3, F4, F5

The Phase 1 slide identifies a gamer concerned with fairness. These arcade design needs are hypotheses for evaluation, not interview findings.

P2 Daniel, 19

Goal: Experience the Odyssey without a confusing first encounter.
Design need: One clear objective per shore, short practice, keyboard or touch controls and literary context.
Flows: F1, F3, F5

The Phase 1 slide identifies a reader with limited technical comfort. The arcade teaching flow is a proposed response to that sketch.

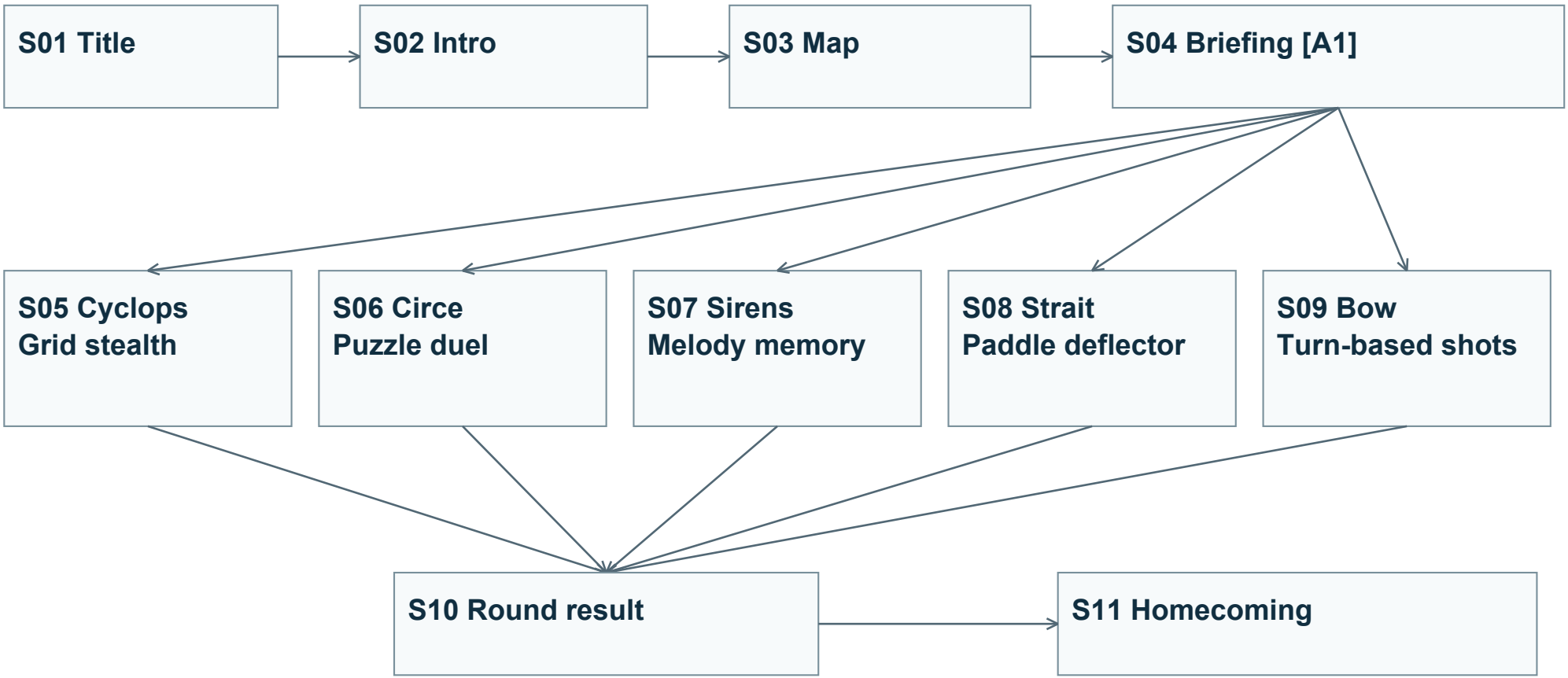
P3 Rosa, 34

Goal: Know what the model changes and keep control over its result.
Design need: Model and local labels, visible strategy review, privacy disclosure, cancellation and a usable fallback.
Flows: F2, F4, F5

AI skepticism and the need for explanations come from the Phase 1 slide. No completed interview or AI trust score is asserted.

Screen hierarchy

S01-S15 / A1 marks the only pretrained-model integration point



Next trial opens its briefing after a clear or assist. The fifth passage opens Homecoming.

Shared overlays: S12 Settings, S13 Pause/leave, S14 Restart, S15 Assisted passage

Content strategy

Screens S01-S05

Screen	Displayed information	Model / storage behavior
S01 Title	Begin/resume, five-trial premise, progress count and Settings.	No model call. Read browser-local save.
S02 Introduction	Odysseus and the voyage, next/skip.	Authored story. Skip preserves the new voyage.
S03 Voyage map	Five shores, cleared/assisted/pending states, locked stops, continue.	No model call. Every earlier shore must be passed.
S04 Encounter briefing [A1]	Rival, myth context, objective, controls, difficulty, mode, practice and preparation.	Only pretrained-model integration point: A1 optional rival planning.
S05 Cyclops round	Grid, sheep, rocks, exits, noise and fixed smash warning, move controls.	Local opponent engine. One evade per warning.

Content strategy

Screens S06-S10

Screen	Displayed information	Model / storage behavior
S06 Circe round	Two puzzle boards, both next pieces, curses and scores, rotate/drop.	Circe plays independently. No model request during a round.
S07 Sirens round	Four numbered/color lanes, Listen/Reply, sequence, Temptation.	Local melody logic targets missed lanes. Optional distinct sound.
S08 Strait round	Shield, six heads, warning paths, rowing control, vortex, hull and time.	Local coordinated attacks. Hold/release input is explicit.
S09 Archery round	12 rings, wind, path preview, three controls, turn and both scores.	Player and rival use the same shot physics. Rival turn disables shooting.
S10 Round result	Clear/loss/practice, score, opponent score where relevant, retry and next.	Completed non-practice rounds update attempts. Failed replays retain past clears.

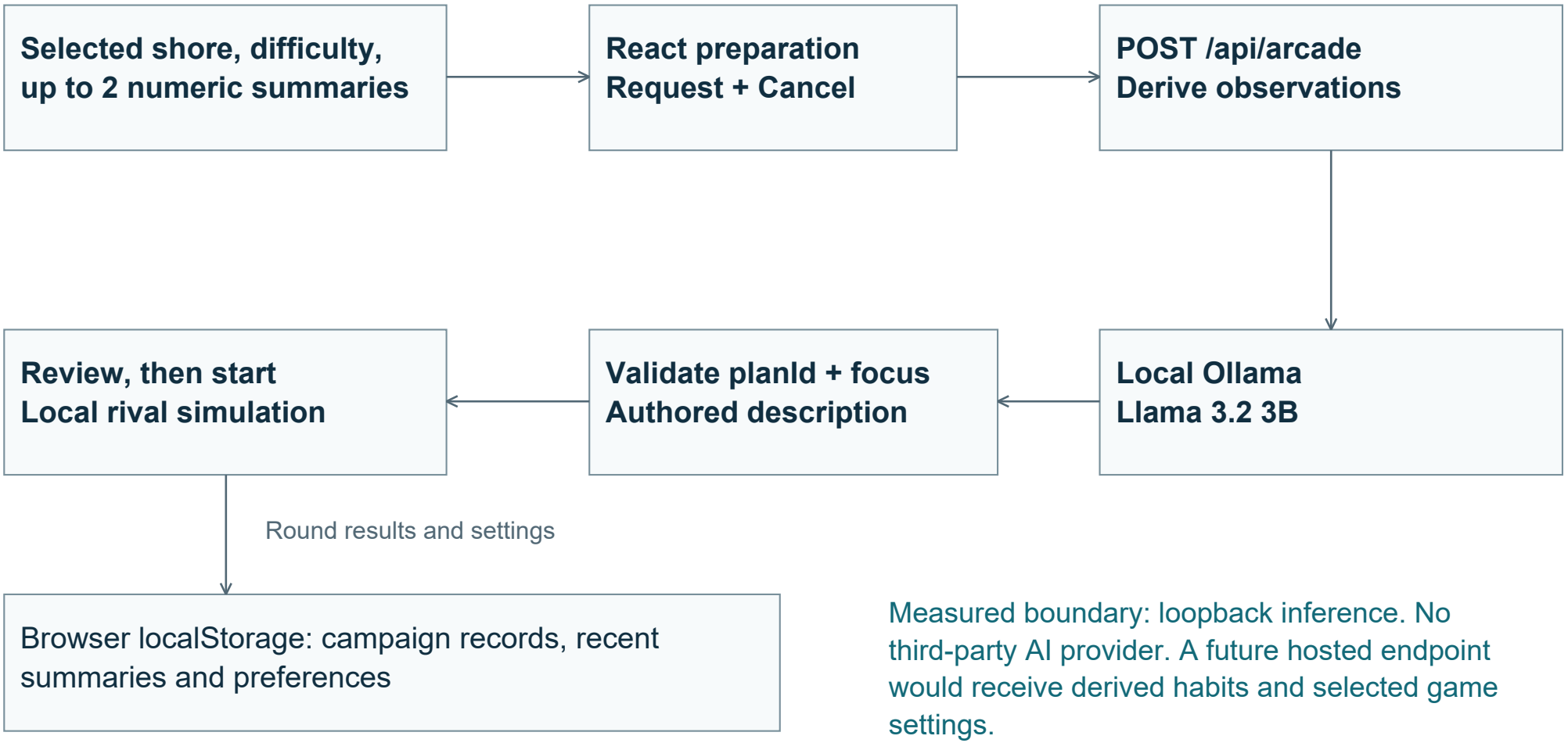
Content strategy

Screens S11-S15

Screen	Displayed information	Model / storage behavior
S11 Homecoming	Family, report of each clear or assisted passage, replay and restart.	No generated ending. Honest saved progress report.
S12 Settings overlay	Sound, still scenery, model connection check, earlier modes.	GET checks availability, not a generated rival strategy.
S13 Pause / leave overlay	Resume, leave warning and return to briefing.	Pause simulation and release held controls. Abandoned round gives no clear.
S14 New voyage confirmation	Explain replacement of current progress, keep or restart.	Replace save only after explicit confirmation.
S15 Assisted passage confirmation	Explain Assisted report label, keep trying or continue with help.	Authored progression choice. Replay can later earn a clear.

Data flow and provider boundary

A1 prepares a policy; local simulation owns the round



Measured boundary: loopback inference. No third-party AI provider. A future hosted endpoint would receive derived habits and selected game settings.

A1 model integration

Optional rival planning, shared by all five encounter briefings

Model

Llama 3.2 3B through local Ollama

Input

Encounter ID, Easy/Normal/Hard and at most two numeric attempt summaries. Server derives habits and losses. Current engines retain zero-valued legacy timing and lane-duration fields. No name, voice or chat text.

Output

JSON with planId and focus. The model chooses one of two adaptive strategies. The game fixes the required focus. Server validates both and supplies an authored description.

Timing

Measured min 7.317s, mean 8.279s, max 11.613s. Use the October 8 six-call experiment. No round runs during preparation. Server timeout 90 seconds, browser limit 95 seconds.

Privacy

Measured setup uses loopback Ollama. No third-party AI provider receives these requests. A hosted endpoint would receive the selected shore, difficulty, derived habits and win count. It would need a changed disclosure.

Player control

Edit difficulty/mode before play, cancel a pending call, regenerate a ready strategy, switch to Solo to reject it, or start a clearly labeled local AI duel after an error.

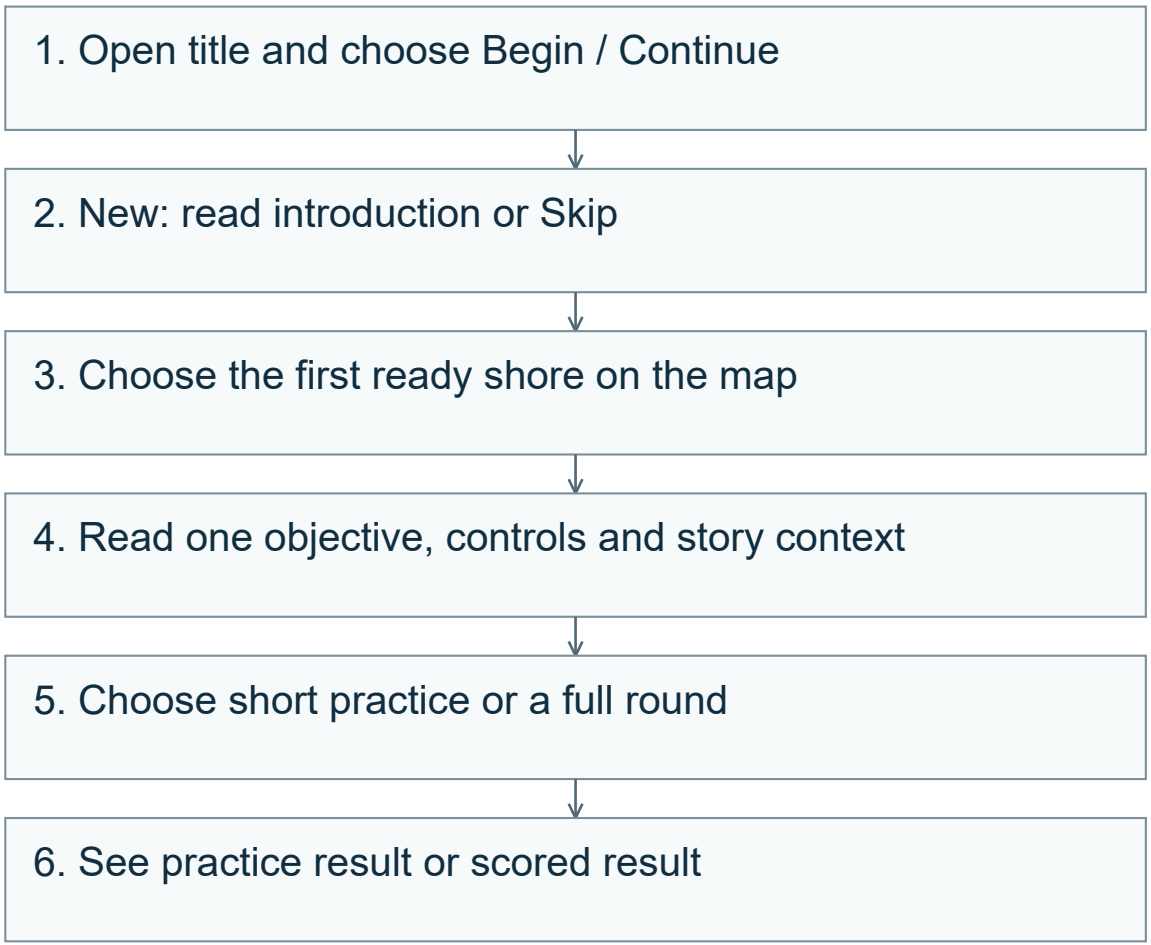
Storage, fixed rules and player control

Preparation changes neither score nor campaign progress

- Current: browser-local save stores cleared/assisted records, attempt counts and at most two recent numeric attempts per shore. Preferences have a separate local key. No account or cloud save.
- Current: a ready model plan stays in memory. Navigating, editing difficulty/mode or cancelling discards it. Starting preparation clears the old candidate. Only a validated result becomes ready.
- Current: collision rules, warnings, difficulty and scoring remain authored. Local opponents act during play. No inference runs inside a real-time match.
- Current: the full-round result commits progress. Practice and an abandoned round award no clear. A failed replay preserves an earlier clear.
- Current: a pending request has elapsed time and Cancel. Generation IDs ignore late replies. Invalid plans expose uncertainty and local fallback choices.
- Proposed: a dedicated Discard strategy action explicitly undoes preparation. A visible privacy detail names loopback inference. The persistent saving warning is already implemented.
- No confidence percentage. A strategy description explains permitted behavior, but does not prove tactical quality. Human evaluation is still required.

F1 Start or resume and learn a trial

P2 Daniel / Implemented



Existing progress

Continue retains clears and preferences. Restart requires S14 confirmation.

Controls are unfamiliar

Practice leaves campaign progress unchanged. Return to the same briefing.

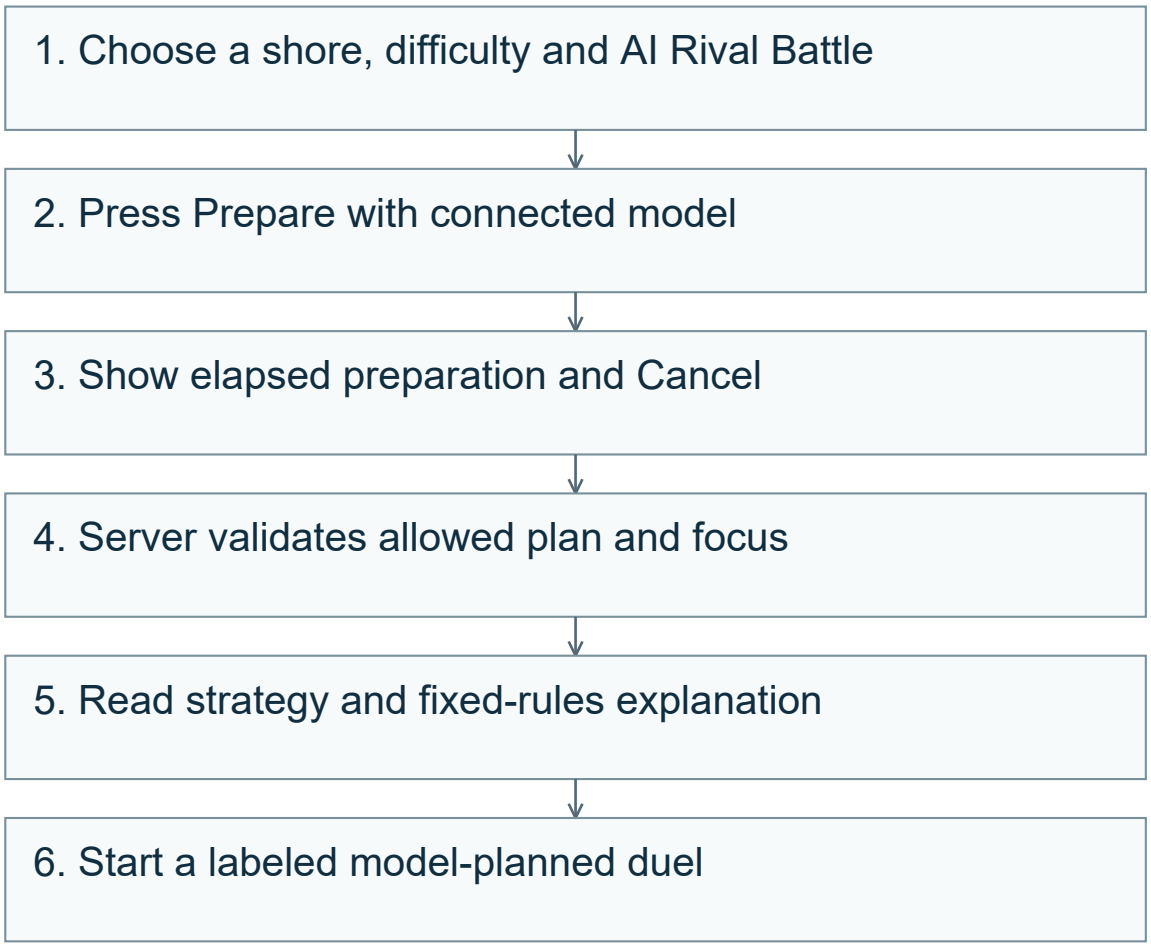
The first usable trial needs no model configuration.

Locked stops explain that earlier trials must be passed. Accessibility cues remain visible without sound.

Screens and states: S01, S02, S03, S04, S05-S09, S10

F2 Prepare, review and accept a model strategy

P1 Maya and P3 Rosa / Implemented, with proposed Discard and privacy detail



Cancel, navigation or edited input

Abort the request and ignore stale replies. Return to initial state. No score or progress changes.

Reject or refine

Prepare another, change difficulty/mode or choose Solo. Proposed Discard removes the ready plan while keeping AI mode.

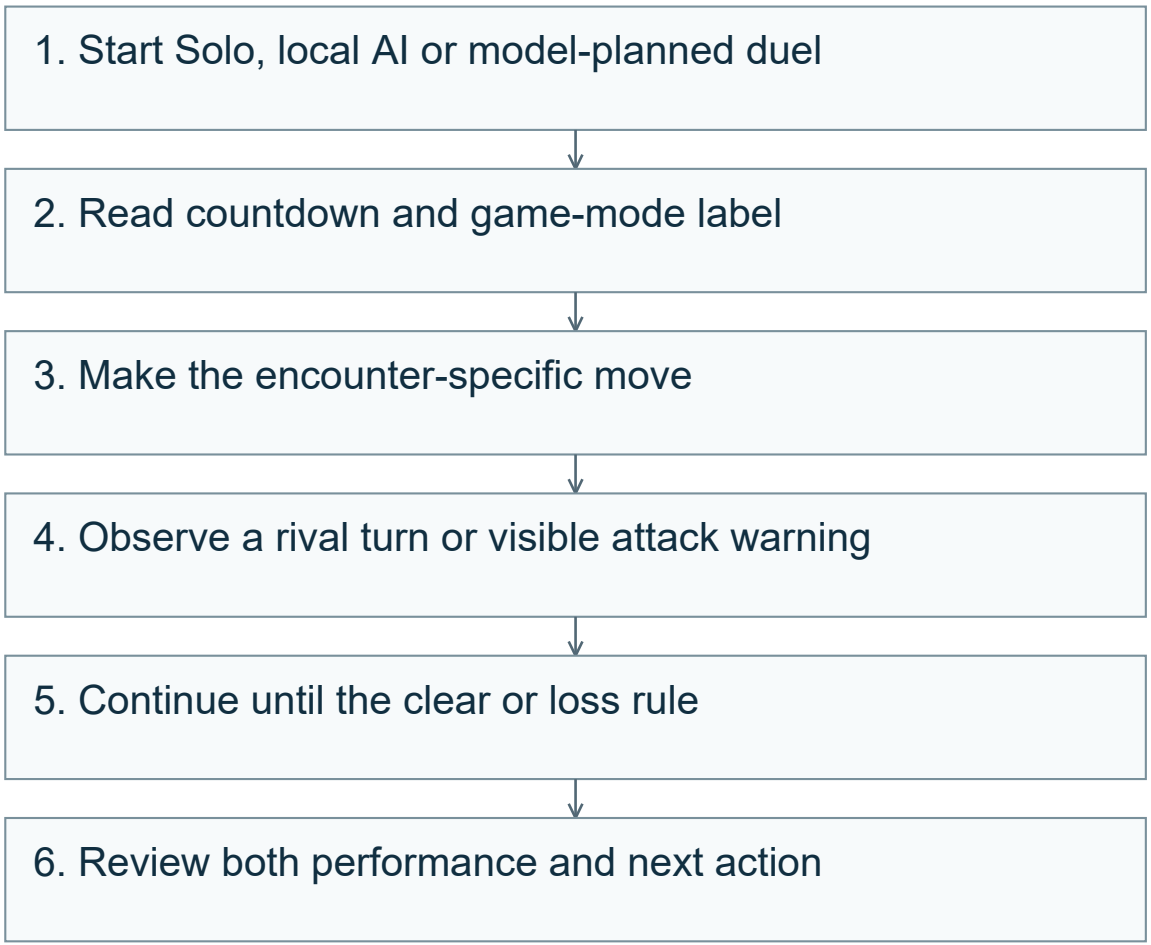
The model chooses a strategy before play. Local heuristic logic performs the rival actions during play.

No confidence percentage. Show the strategy, its limits and user controls before committing to the round.

Screens and states: S04 / A1 initial, loading, success, uncertain, error

F3 Play against an active mythic rival

P1 Maya and P2 Daniel / Implemented



Pause / hidden tab / leave

Pause clocks and release held keys. Resume the same round, or confirm leaving with no awarded clear.

Loss

Try again, practice, inspect the briefing or request an assisted passage. Earlier clears remain.

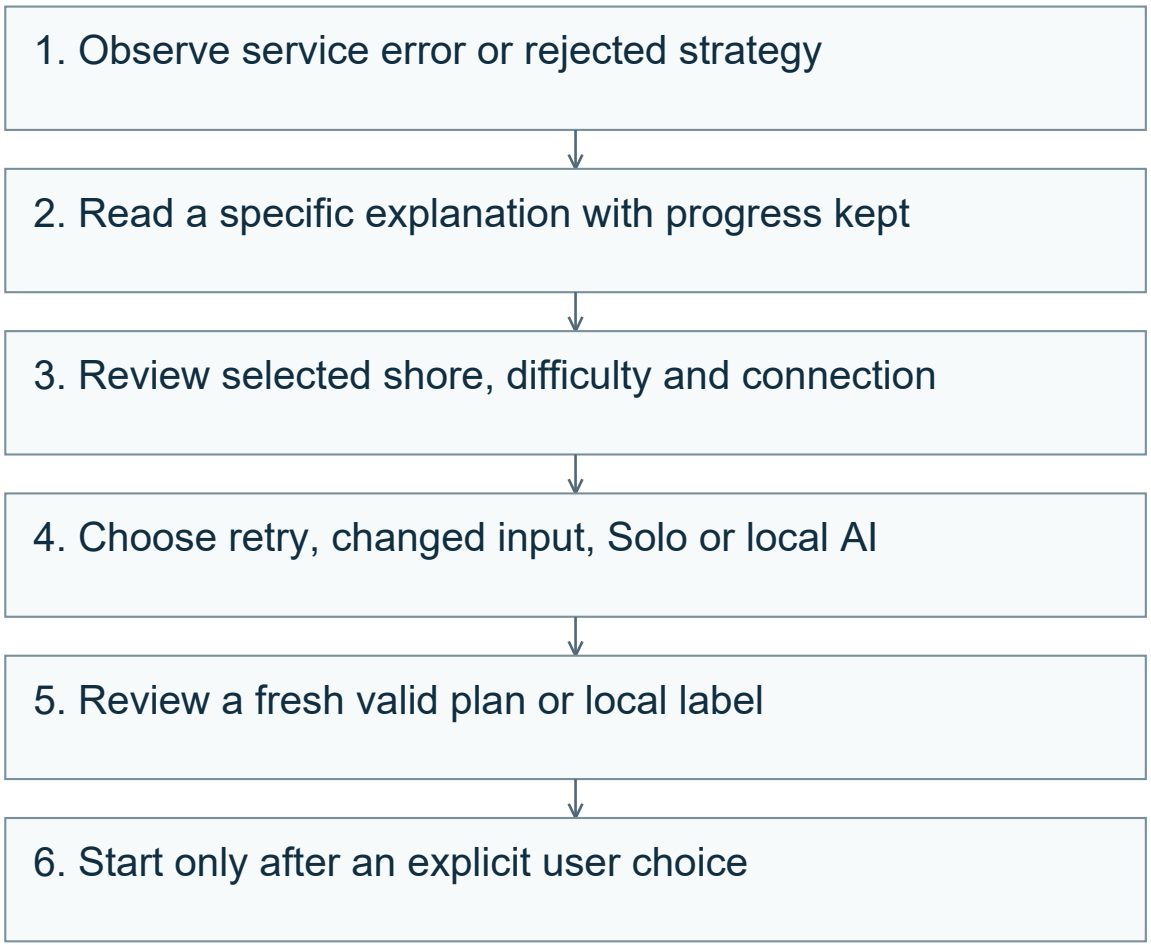
Cyclops, Circe, Sirens, strait and bow use distinct controls. See W11-W17 for their playable states.

Scores and collision rules come from authored code. The model cannot shorten warnings or declare a win.

Screens and states: S05-S09, S13, S10

F4 Recover from a failed or unsuitable AI plan

P3 Rosa and P1 Maya / Implemented



Poor / empty / unsupported output

Invalid JSON, extra fields, illegal plan or wrong focus is rejected. No match starts. For a legal but unwanted tactic, regenerate or reject.

Offline / timeout / rate limit

Retry when ready, cancel a long wait, or choose Solo/local AI. Provider 429 maps to a busy-service message.

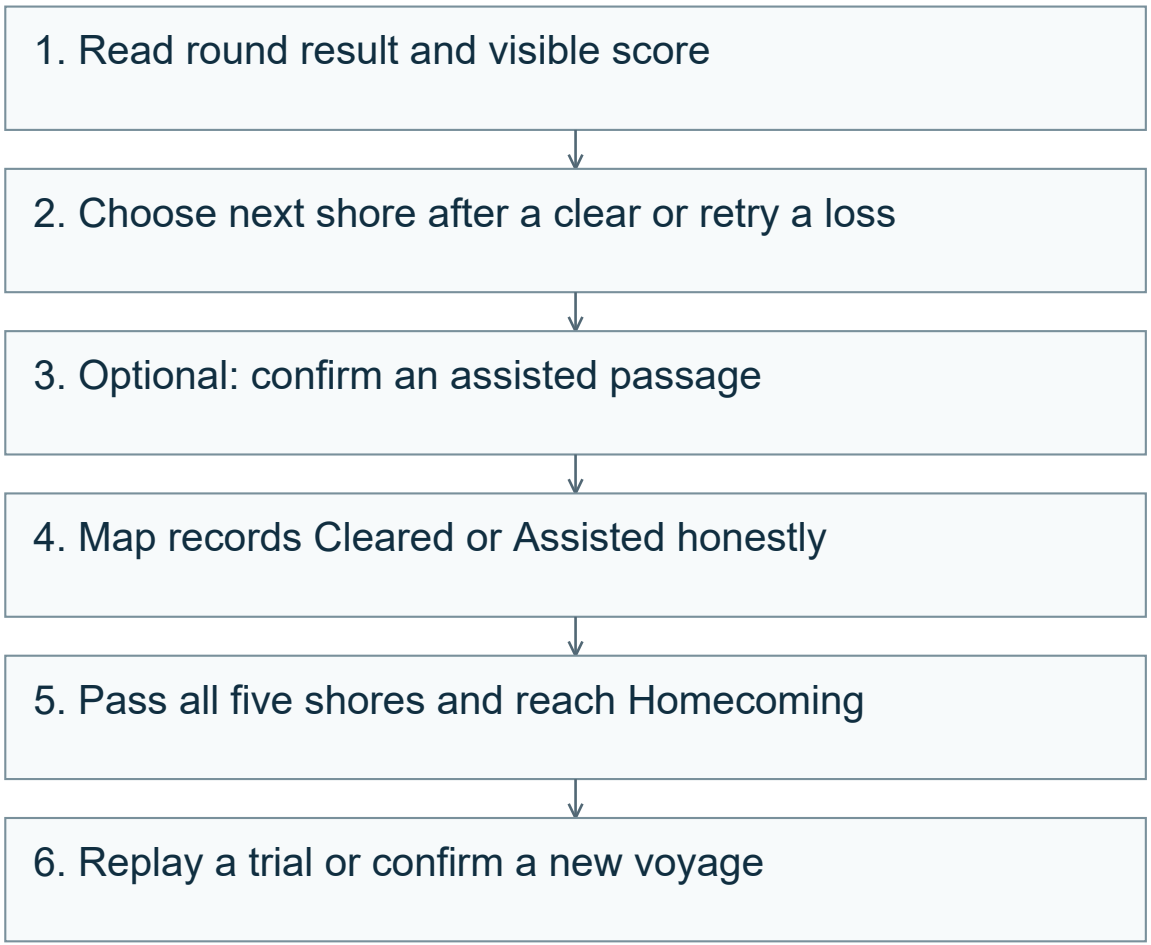
The live spike triggers real missing-model and malformed-request failures. 429 behavior is designed and unit-tested, not live-induced.

Unsupported difficulty and oversized/invalid requests stop before model contact. Do not silently replace a failed model response.

Screens and states: S04 / A1 uncertain, error, invalid input; S12

F5 Continue, request assistance and finish

P2 Daniel, P1 Maya and P3 Rosa / Implemented



Reconsider help

Keep trying closes S15 without changing progress. A later successful replay upgrades Assisted to Cleared.

New voyage

S14 explains that restarting replaces progress. Keep voyage returns unchanged.

Assistance improves access to the narrative without claiming a skill-based clear.

Practice and abandoned rounds award no campaign progress. Replays preserve earlier success even when the player loses.

Screens and states: S10, S15, S03, S11, S14

Edge cases and recovery

EC1-EC3

Scenario / states	Recovery	Observed vs designed
EC1 Unavailable / timeout / 429 F4, A1 error	Keep progress. Retry, Cancel, Solo or a labeled local AI duel. Preparation never consumes game time.	E1 observes real provider 404. 90s timeout and 429 mapping exist; those cases were not induced live.
EC2 Poor, wrong or empty output F2/F4, A1 uncertain	Reject invalid JSON or unsupported plan/focus. Regenerate or change inputs. Legal but unwanted tactics remain rejectable.	Strict schema and server validation; fresh spike measures legal outputs, not tactical enjoyment.
EC3 Invalid / unsupported input F4, S04 validation	Explain supported choices and keep valid settings. Reject invalid input before model contact. Inline field guidance is proposed.	E3 records actual application 400 for impossible difficulty.

Edge cases and recovery

EC4-EC6

Scenario / states	Recovery	Observed vs designed
EC4 Cancel or late reply F2, A1 loading	Abort and increase request generation ID. Ignore old replies after cancel, input edit, navigation or regeneration.	Implemented request token and AbortController. No clear or score can arrive from a model reply.
EC5 Inappropriate / biased output F2/F4, A1 uncertain	Allow permitted strategy IDs. Show authored descriptions. Reject generated prose and all extra fields.	This limits the output surface; fair tactical behavior still needs player evaluation.
EC6 Hidden tab / lost held input F3, S13 pause	Pause simulation and release rowing/paddle controls. Resume explicitly. Explain that reload restarts the round.	Implemented focus/visibility handling. Completed progress stays local.

Edge cases and recovery

EC7-EC8

Scenario / states	Recovery	Observed vs designed
EC7 Local storage unavailable F1/F5, S01/S03	Show the implemented persistent saving warning. Keep the in-memory voyage and explain that reload may lose progress.	Storage exceptions trigger a persistent status notice. This behavior is implemented.
EC8 Migrated old save / abandoned round F1/F5, S03/S10	Retain old clears, insert Circe/strait pending. Clear earlier shores before replaying later ones. No clear for an unfinished round.	Save schema 2 migration. Assisted and practice labels remain explicit.

AI state and screen coverage

28 wireframes cover every screen and major interaction state

A1 state	Wireframe	Flow	Expected action
Initial	W05	F2	Prepare model or start local play
Loading	W06	F2/F4	Elapsed time / Cancel / Practice
Success	W07	F2	Review / Start / regenerate / edit
Uncertain	W08	F2/F4	Reject unsupported plan / retry / local
Error	W09	F4	Retry / check connection / local

Screen coverage

S01 W01-W02; S02 W03; S03 W04/W28; S04 W05-W10/W27; S05 W11; S06 W12; S07 W13-W14; S08 W15; S09 W16-W17; S10 W18-W20; S11 W21; S12 W22; S13 W23-W24; S14 W25; S15 W26.

W10 inline validation and W27 explicit Discard are proposed. W28 saving warning is implemented. The current server already rejects invalid requests.

Low-fidelity wireframes

W01 and W02

W01 Title - new player

S01 / empty / F1

Voyage artwork placeholder

Farshore / The Odyssey: Arcade Trials

Five distinct trials. Keyboard or touch.

Begin voyage / Settings

Implemented

A single primary action starts the game without AI setup.

W02 Title - returning player

S01 / saved progress / F1/F5

Voyage artwork placeholder

Continue voyage / 2 of 5 cleared

Continue voyage

New voyage / Settings

Implemented

Continue reads browser-local progress. New voyage opens S14.

Low-fidelity wireframes

W03 and W04

W03 Introduction

S02 / story / F1

- Odysseus portrait placeholder
- Short authored Odyssey context
- Introduction progress
- Next / Skip

Implemented

Skip moves to the map. This story text is authored, with no generation.

W04 Five-stop voyage map

S03 / progress / F1/F5

- Your way home / 2 of 5 passed
- 1 Cyclops, 2 Circe, 3 Sirens, 4 Strait, 5 Bow
- Each stop: Cleared / Assisted / Ready / Locked
- Continue voyage / choose an unlocked stop

Implemented

Locked later trials explain the preceding requirement. Migration adds two new pending shores.

Low-fidelity wireframes

W05 and W06

W05 Model planning - initial

S04 / A1 initial / F2

Rival, objective, controls, myth context

Difficulty: Easy / Normal / Hard

Arcade Solo / AI Rival Battle

Local adaptive AI is ready now

Prepare with connected model

Start local AI duel / Practice

Implemented; privacy detail Proposed

A1 begins only on Prepare. Numeric attempt summaries provide context. Proposed detail: visible loopback-provider disclosure.

W06 Model planning - loading

S04 / A1 loading / F2/F4

Preparing Circe's strategy...

12 seconds elapsed. No round clock is running.

Input settings remain selected

Cancel preparation

Practice (cancels preparation before play)

Implemented

Clear feedback appears immediately. Cancel ignores stale replies. Start is disabled while loading.

Low-fidelity wireframes

W07 and W08

W07 Model planning - ready

S04 / A1 success / F2

Model strategy ready: Chain Alchemist

Circe previews a piece to build chains.

Normal rules stay fixed. Rival acts locally.

Start model-planned duel

Prepare another / edit difficulty / Solo

Implemented

Review precedes play. Changing mode or difficulty discards the ready plan. No scoring changes during preparation.

W08 Model planning - unsupported

S04 / A1 uncertain / F2/F4

The model returned an unsupported strategy.

Nothing changed. No game has started.

No confidence percentage

Retry model preparation

Start local AI duel / Solo / Practice

Implemented

Invalid or extra fields cannot become a rival plan. Uncertain describes suitability, not a measured probability.

Low-fidelity wireframes

W09 and W10

W09 Model planning - service failure

S04 / A1 error / F4

- The model is unreachable or took too long.
- Your voyage is kept.
- Retry model preparation / check Settings
- Start local AI duel / switch to Solo
- Practice / return to map

Implemented

A failed request never silently becomes model-planned play. Every fallback has its own label.

W10 Unsupported input

S04 / validation error / F4

- Choose a supported encounter and difficulty.
- Restore a supported choice: Easy / Normal / Hard
- At most two numeric summaries are accepted.
- Try again / return to map

Server validation implemented; inline state Proposed

The visible selectors constrain ordinary input. The server rejects malformed or tampered requests before inference.

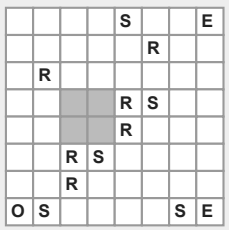
Low-fidelity wireframes

W11 and W12

W11 Polyphemus grid

S05 / player / smash warning / F3

Grid stealth / turn / hits / current rival mode



O = Odysseus
S = sheep / R = rock
E = exit
Gray = warned smash

Marked 2 x 2 smash area. One evade left.

Up / Left / Right / Down / Sneak-Dash / Hide

Pause

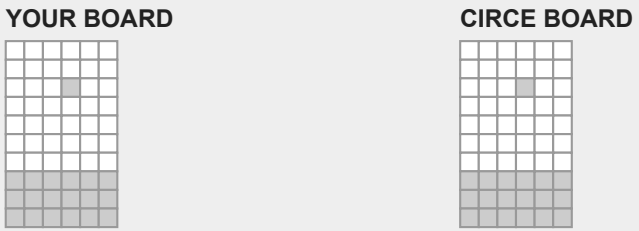
Implemented

Blind rival uses noise clues. The visible target locks before the single evade. Exits and sheep have symbols.

W12 Circe puzzle duel

S06 / simultaneous play / F3

Your score / Circe score / time / rival mode



Your next triplet / Circe next triplet / curse bank

Choose column / Rotate / Drop

Pause

Implemented

Two boards make the opponent's moves visible. Moly matches purify swine and send curses.

Low-fidelity wireframes

W13 and W14

W13 Sirens melody

S07 / Listen / F3

Listen / melody 2 of 5 / Temptation 20%

A / 1	S / 2	D / 3	F / 4

Highlight and optional tone play the sequence

Wait for Reply before entering notes

Pause

Implemented

Numbered pads and phase text work without audio. Trick rests appear visibly.

W14 Sirens counter melody

S07 / Reply / F3

Reply / entered notes / remaining time

A / 1	S / 2	D / 3	F / 4

Temptation increases on wrong or missing notes

Press pads or use keys / Pause

Implemented

Local Siren logic targets misses within the current round. Previous-round misses inform model policy selection, not initial lane targeting. Pads also have labels.

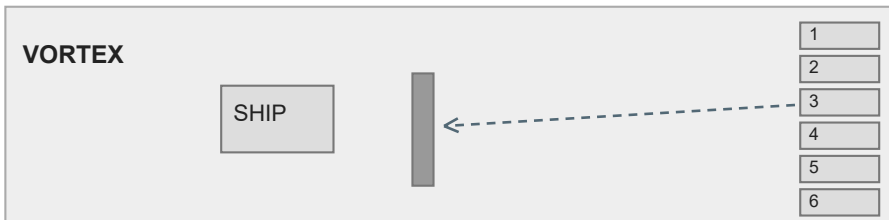
Low-fidelity wireframes

W15 and W16

W15 Twin-paddle escape

S08 / active strikes / F3

Hull 5 / survival time / rival mode



Telegraphed strike paths and visible pull

Shield Up/Down or drag / hold Row or Space

Pause

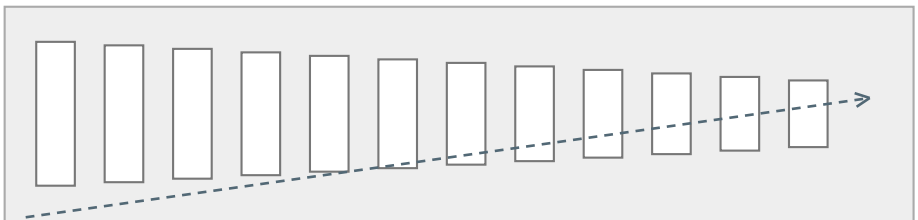
Implemented

Release Row to stop rowing. Pause or focus loss releases every held input.

W16 Archery - player turn

S09 / aim and release / F3

Your turn / shot 1 of 5 / both ring totals



Wind direction/speed and predicted arrow path

Aim / elevation / tension sliders

Release arrow / Pause

Implemented

Prediction uses the same physics as the shot. Inputs identify their values and can use keyboard controls.

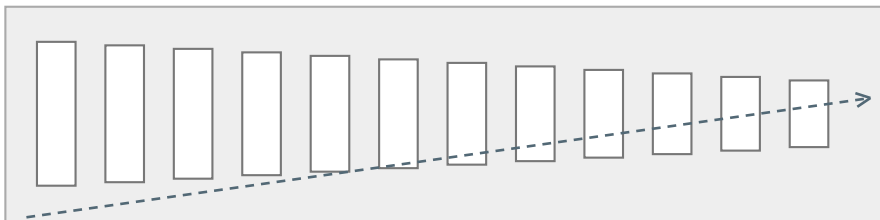
Low-fidelity wireframes

W17 and W18

W17 Archery - rival turn

S09 / opponent shot / F3

Antinous' turn / shooting controls disabled



Wind Reader or temporary Block-shot action

Next wind appears when player turn resumes

Pause

Implemented

The rival takes an independent scored turn. A temporary block is visible and expires after one player shot.

W18 Round cleared

S10 / success / F3/F5

Encounter cleared / score / difficulty / mode

Both scores for Circe and Antinous

What the player achieved

Sail to next trial / Practice again / Map

Implemented

Update progress once after a full completed round. The final bow clear goes to Homecoming.

Low-fidelity wireframes

W19 and W20

W19 Round lost

S10 / retry / F3/F5

- Another chance / score and failure explanation
- Your earlier clears are kept
- Try again / Practice / Map
- Use an assisted passage

Implemented

Loss keeps a usable path forward. Assisted passage requires S15 confirmation.

W20 Practice result

S10 / practice complete / F1/F3

- Practice complete / Ready for the full duel?
- Practice changes no voyage progress
- Back to challenge / Practice again / Map

Implemented

Practice has a separate result label and awards no campaign clear.

Low-fidelity wireframes

W21 and W22

W21 Homecoming report

S11 / campaign complete / F5

- Odysseus, Penelope and Telemachus placeholders
- Home is more than a shore
- Five-trial report: Cleared or Assisted
- Replay a trial / New voyage

Implemented

The ending is authored. The report shows each shore's current Cleared or Assisted status.

W22 Voyage settings

S12 / overlay / F1/F4

- Music & sounds / Still scenery
- AI connection: available / unavailable / checking
- Check connection again / Close
- Both modes work without model preparation
- Earlier mystery or advice mode links

Implemented

A catalog check is separate from generating a strategy. Closing returns focus to the main heading.

Low-fidelity wireframes

W23 and W24

W23 Paused round

S13 / pause / F3

Paused. Every game clock has stopped.

Current round remains in memory

Resume / leave round

Hidden tab or lost focus also pauses

Implemented

Resuming does not replay held inputs. The round is not saved across browser reloads.

W24 Leave round confirmation

S13 / leave / F3

Leave this round?

You will return to the briefing. No clear is awarded.

Keep playing / Leave round

Implemented

Protect against accidental loss of an in-progress attempt. Completed voyage progress remains.

Low-fidelity wireframes

W25 and W26

W25 New voyage confirmation

S14 / restart / F1/F5

Start a new voyage?

This replaces the current voyage progress.

Keep voyage / Start again

Implemented

Only Start again replaces the save. Keep voyage returns unchanged.

W26 Assisted passage

S15 / help decision / F5

A little help from Athena

Continue the story. This shore will say Assisted.

Replay later to earn a clear

Keep trying / Continue with help

Implemented

Assistance describes exactly how the homecoming report changes.

Low-fidelity wireframes

W27 and W28

W27 Discard a prepared plan

S04 / A1 success / undo preparation / F2

Prepared strategy and explanation

Start / Prepare another / Discard strategy

Discard returns to local AI without a match

Edit difficulty or mode

Proposed

Proposed Discard is an explicit undo of preparation. It clears the unplayed plan. Current edit/mode controls already clear it.

W28 Saving unavailable

S03 / storage error / F1/F5

Saving is unavailable on this browser.

You can continue, but reload may lose progress.

Current in-memory voyage map

Keep playing / Settings

Implemented

A persistent warning makes local storage failure visible while keeping play available.

AI spike for the arcade trials

Six sequential actual Llama calls on October 8, 2026

Minimum 7.317s Mean 8.279s Maximum 11.613s

Sample	Seconds	Result	Analytic review
S1 First-time Cyclops	11.613	focused/center	Legal focused/center. Differs from advisory sweep preference. No optimality claim.
S2 Circe after two losses	7.597	chain/balanced	Legal chain/balanced. Matches repeated-loss preference.
S3 Sirens with red-lane misses	7.317	echo/balanced	Legal echo/balanced. Prior red misses reach the prompt. Local targeting starts fresh each round.
S4 Strait after one hull loss	7.889	pincer/balanced	Legal pincer/balanced. Differs from advisory undertow preference.
S5 Archery after low ring score	7.614	feint/balanced	Legal feint/balanced. Fairness needs a player study.
S6 Cyclops with left-side habit	7.641	focused/left	Legal focused/left. Required habit focus is preserved.

All six outputs were legal. S1 and S4 differ from advisory prompt preferences. This finite catalog limits output risk, but legal results do not establish fair or enjoyable play.

Failure evidence and loading decisions

Separate actual provider responses from app validation and unmeasured cases

Probe	Boundary	Observed output
E1 Missing model	Real provider	HTTP 404, model not found
E2 Malformed JSON	Real provider	HTTP 400, unexpected EOF
E3 Unsupported difficulty	App before provider	HTTP 400, choose supported encounter and difficulty

- Display a preparation message immediately, show elapsed seconds and offer Cancel. The observed maximum exceeds ten seconds. No gameplay clock runs during the wait.
- Reject unsupported results before a match. Retain selections/progress and provide explicit Solo or local AI choices. Check connection, retry or leave without a dead end.
- No live 429 or sustained quota threshold was measured. Local inference used no billed API. Hardware, electricity and future hosting cost are unmeasured.
- Standalone script captures raw provider request/body and final adapter results. Temperature 0, strict JSON schema, 100 output-token cap and no streaming. No repeated, randomized or user-study benchmark.

Sources and remaining human evidence

Traceable design evidence and honest research status

Professor's Phase 2 brief

original/CMPS_4200_HCI_Phase_2_Project_Requirements.pdf

Phase 1 proposal

original/Farshore_Phase12nd.pptx

Current prototype

src/ArcadeApp.tsx; src/lib/arcade.ts; src/lib/mythic.ts; server/arcade.ts

October 8 spike

evidence/phase2-arcade-spike-2026-10-08.json

Ollama structured outputs

https://docs.ollama.com/capabilities/structured-outputs

Ollama compatibility

https://docs.ollama.com/api/openai-compatibility

Literary source

public/story-sources.md

Add actual peer critique and interview records if available. Review the contribution statement before submission. No invented reviewer feedback, human study results or instructor approval is included.